

Measuring Anonymity in Complex Networks

Rachel G. de Jong, Mark P. J. van der Loo, Frank W. Takes | 6-7-2021



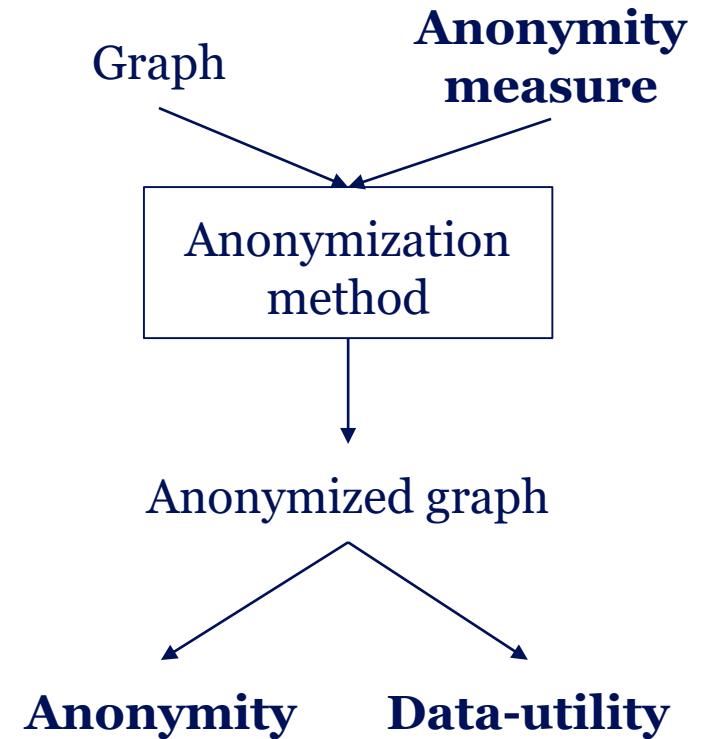
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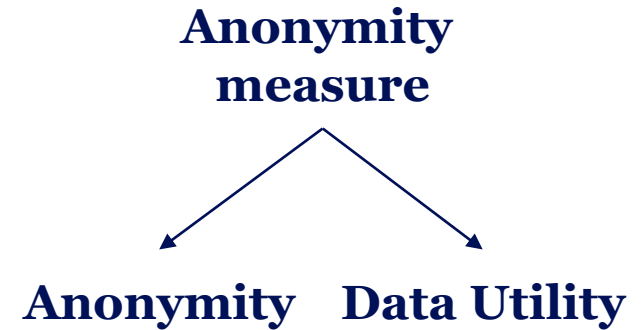
Introduction

- Statistics Netherlands (CBS) created network about the Dutch population [1] from register data
 - Want to share this network -> Need to preserve privacy
 - Common for tabular data -> Statistical Disclosure Control [2]
 - Graph Data:
 - Need to take into account *structural position* of vertices
 - **Measuring Anonymity:**
 - Given a graph, how anonymous are the entities?
 - Frequent next step: **Anonymization**
 - Adjust the graph in order to preserve privacy
 - Trade-off: anonymity vs. data utility
- > **Our goal:** Develop an anonymity measure parametrized in the anonymity / utility trade-off



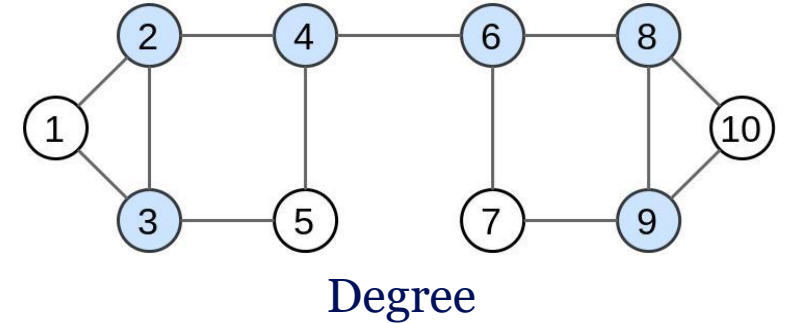
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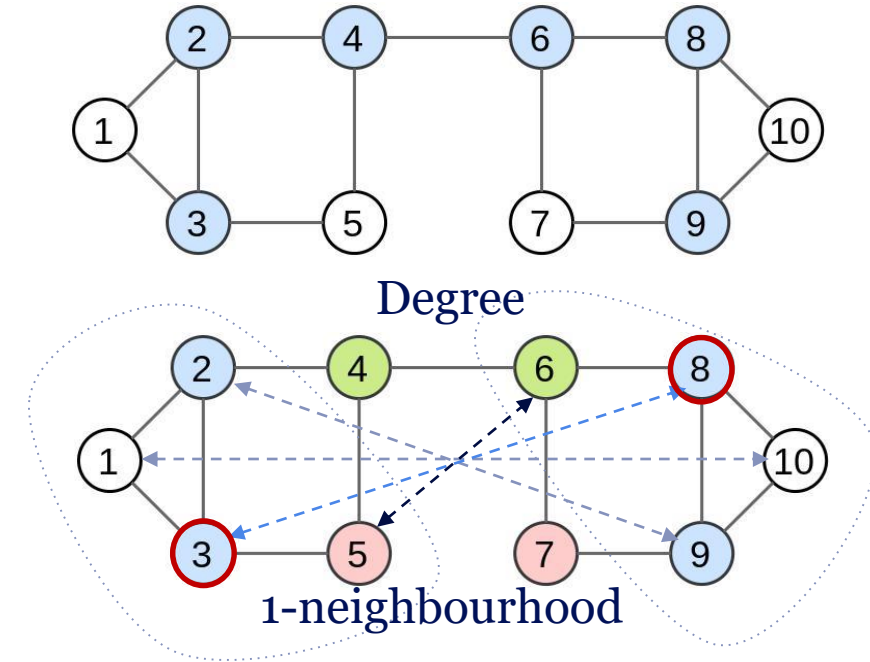
Related Work: k-Anonymity

- Various approaches [3]; amongst which k-anonymity
 - For each vertex, there should be $k-1$ equivalent vertices
 - > same **equivalence class**, size corresponds to **anonymity**
 - Equivalent with respect to:
 - **Degree** [4]
 - Equivalent if degrees are equal
 - **1-Neighbourhood** [5]
 - **Automorphism** [6]



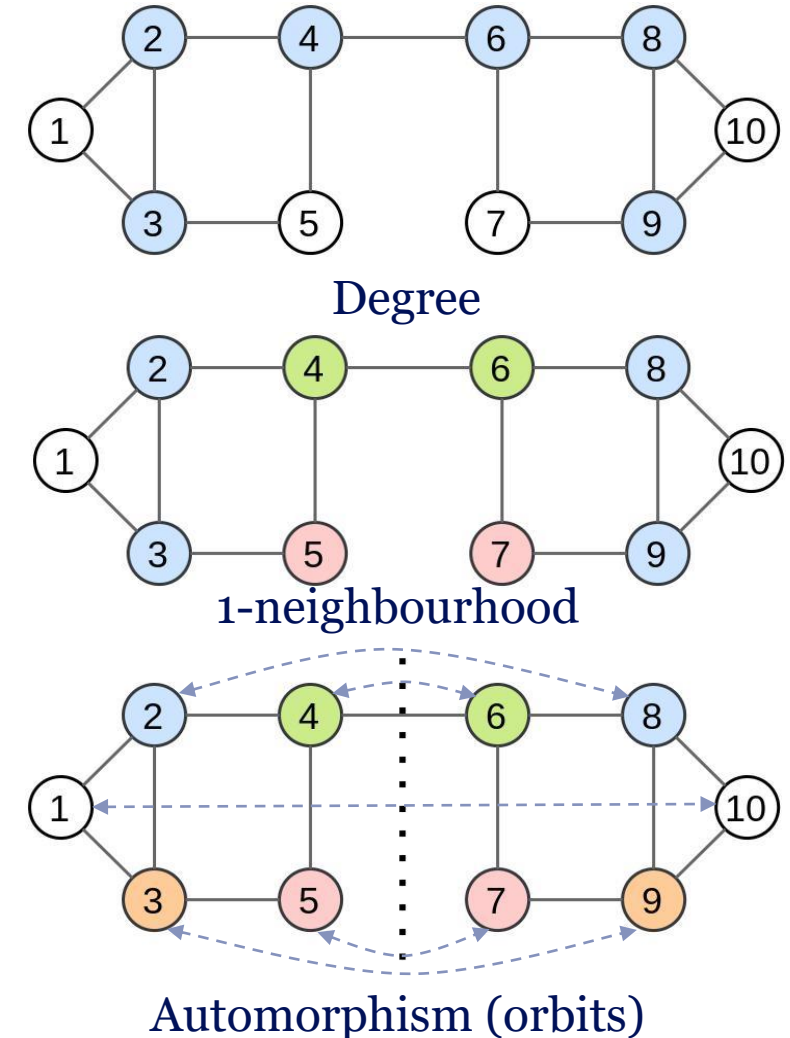
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 - Equivalent if 1-neighbourhoods are isomorphic
 - > Degree + triangles
 - **Automorphism** [6]



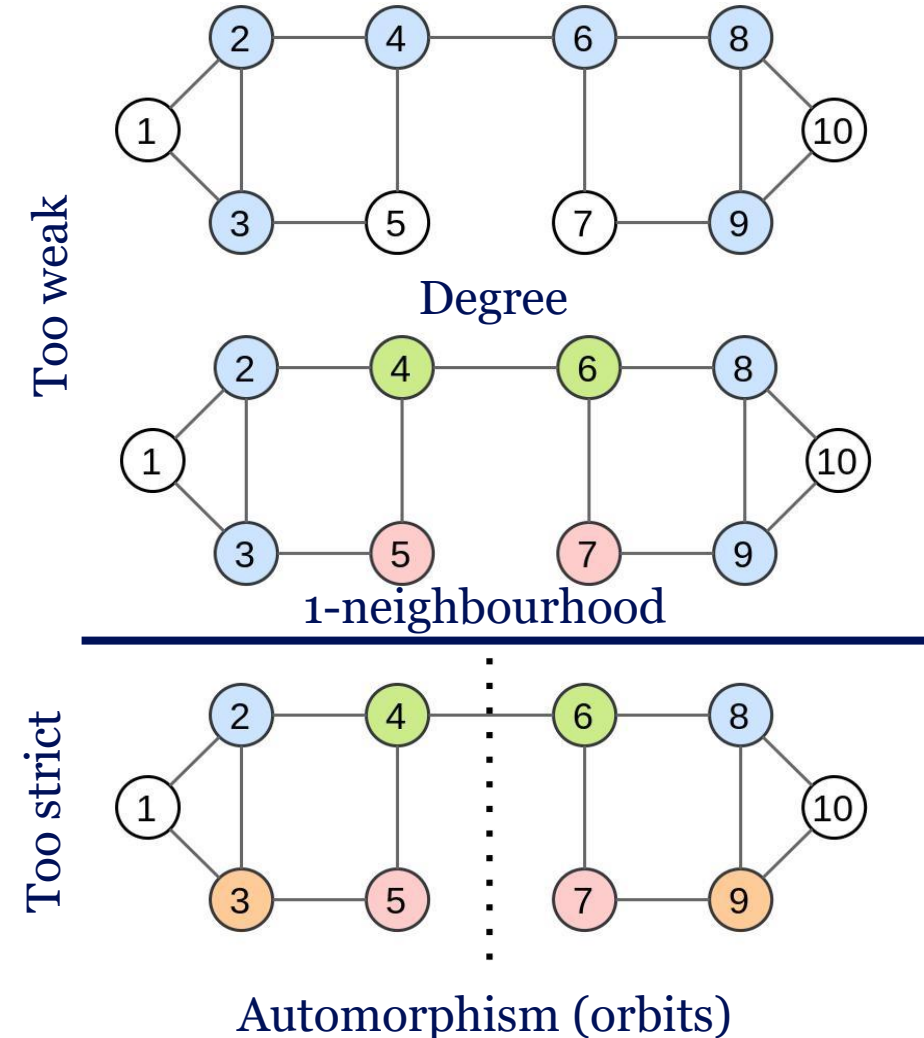
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 - Automorphism: isomorphism from the graph onto itself
 - Equivalent vertices are in same **orbit**: at least one isomorphism mapping the vertices onto each other
 - Perfect measure: vertices in same orbit are **indistinguishable**
 - Expects **symmetry** in graph



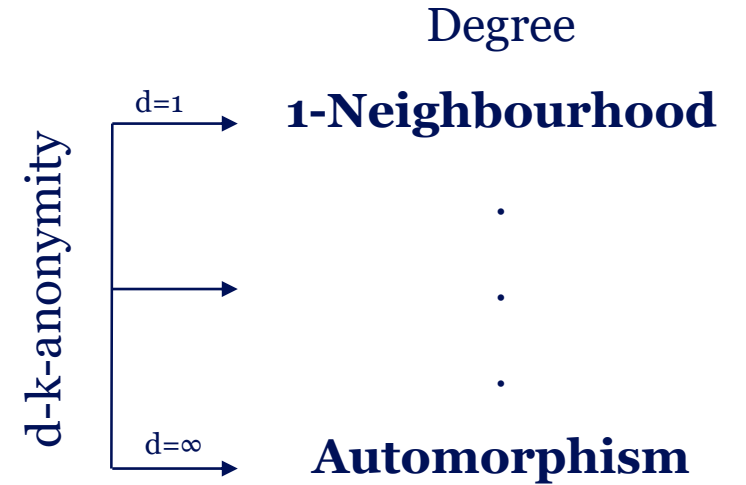
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Approach: d-k-Anonymity

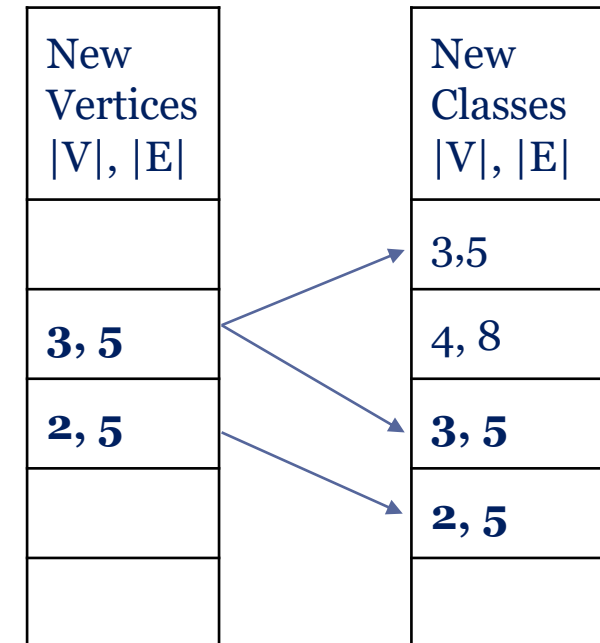
- **Two vertices are equivalent iff:**
 1. Their respective d-neighbourhoods are isomorphic
 2. At least one isomorphic function maps v_1 onto v_2 (orbit)
- Same as **1-neighbourhood** if $d = 1$
- Same as **k-automorphism** if $d \geq \text{diameter}$
 - Diameter: longest shortest path in graph
- **Same structural position in d-neighbourhood**



Implementation: d-k-Anonymity

- Implemented in C++ using the **Nauty** framework [7]
 - Check isomorphism by **canonical labeling**
- Iteratively increase neighbourhood ($d = 1, 2, \dots, 5$)
- **Heuristics:** reduce candidate classes by choosing a set of *feasible vertices*
 - Same number of vertices and edges in d-neighbourhood
 - Same degree distribution in d-neighbourhood (useful in networks with limited difference in degrees e.g. Family Network)

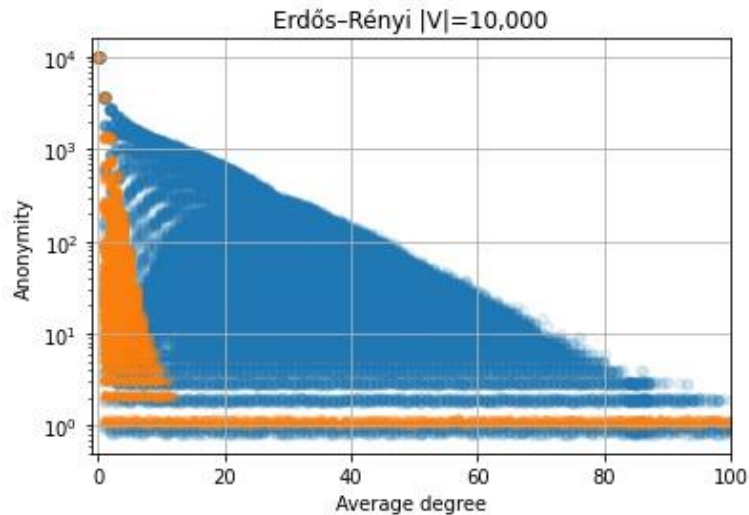
	Vertices	Runtime
Family network	15.7 million	~2 hr
Erdős Rényi	10,000	< 70.0 sec
Barabási Albert	10,000	< 17.5 sec



Results

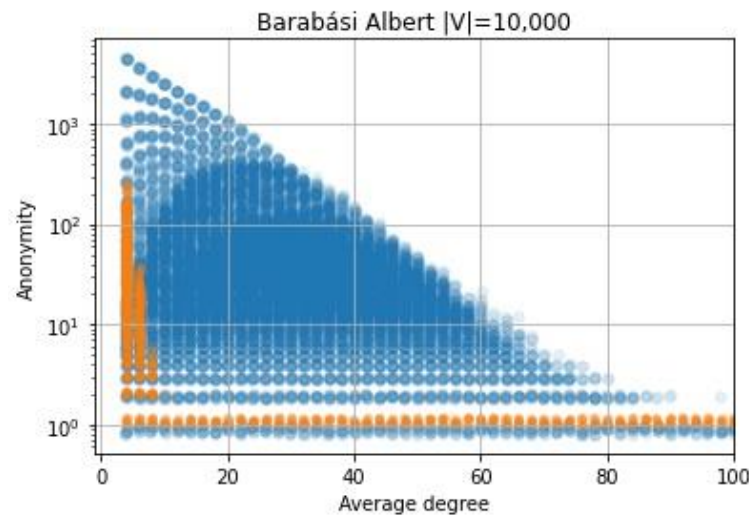
- **Erdős Rényi¹ [8]**

- Random graph, randomly select edges



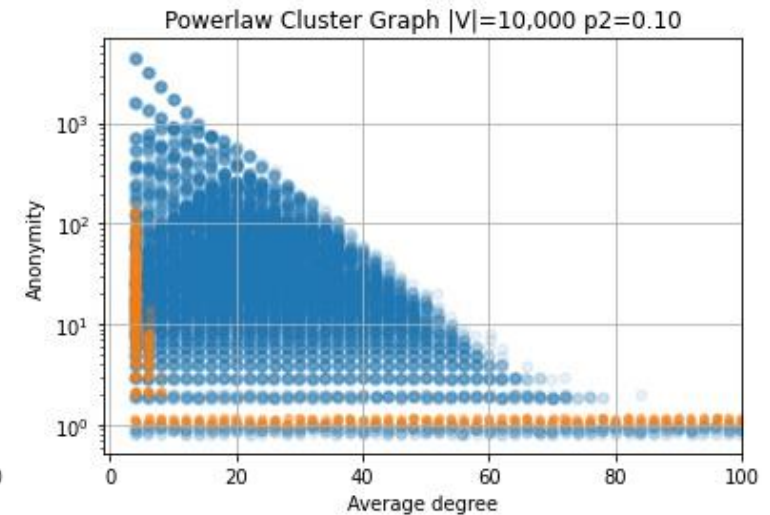
- **Barabási Albert¹ [9]**

- Preferential attachment
- Growth



- **Powerlaw Cluster Graph² [10]**

- BA + clustering step: after each edge, add a triangle with probability p_2



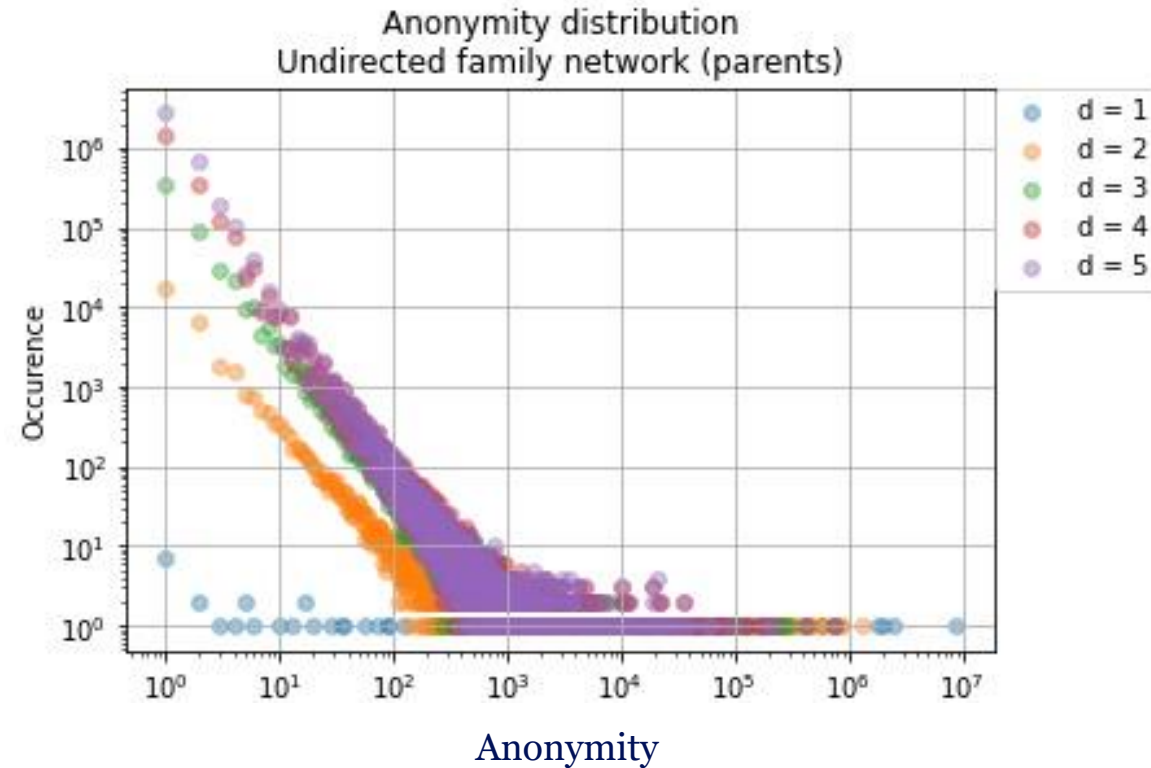
d = 1
d = 2

Generated using ¹igraph [11] ²networkx [12]

Family Network

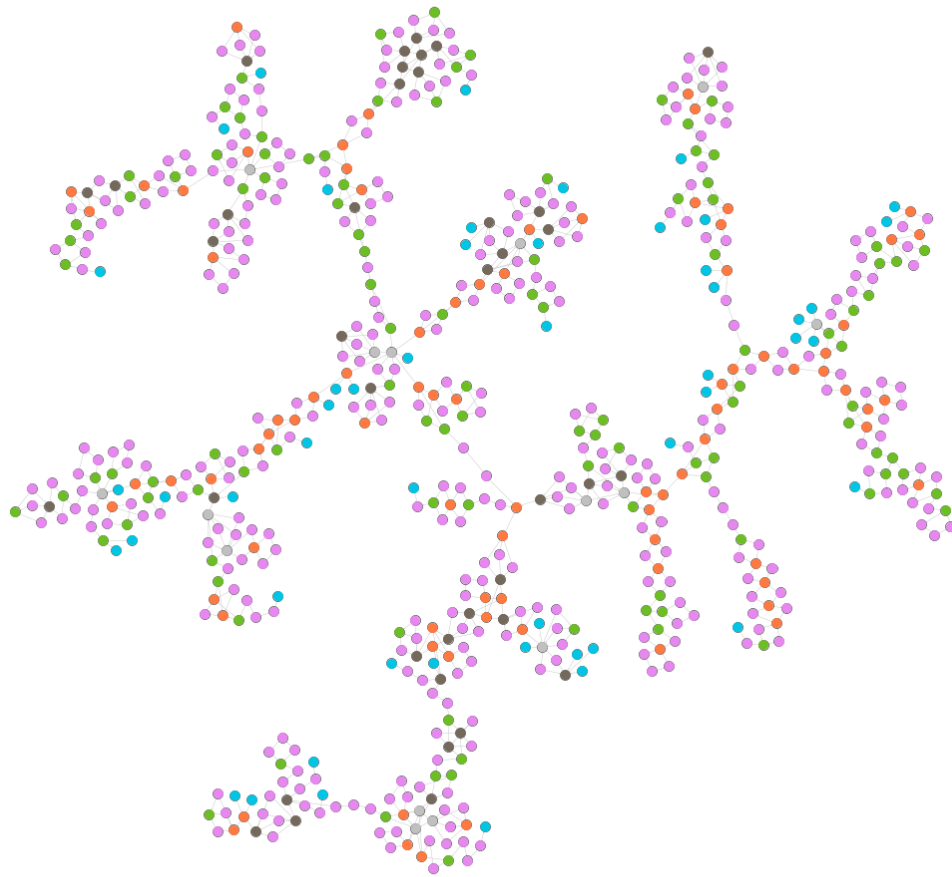
Parent child relations

- $|V| = 15.7$ million
- $|E| = 19.1$ million

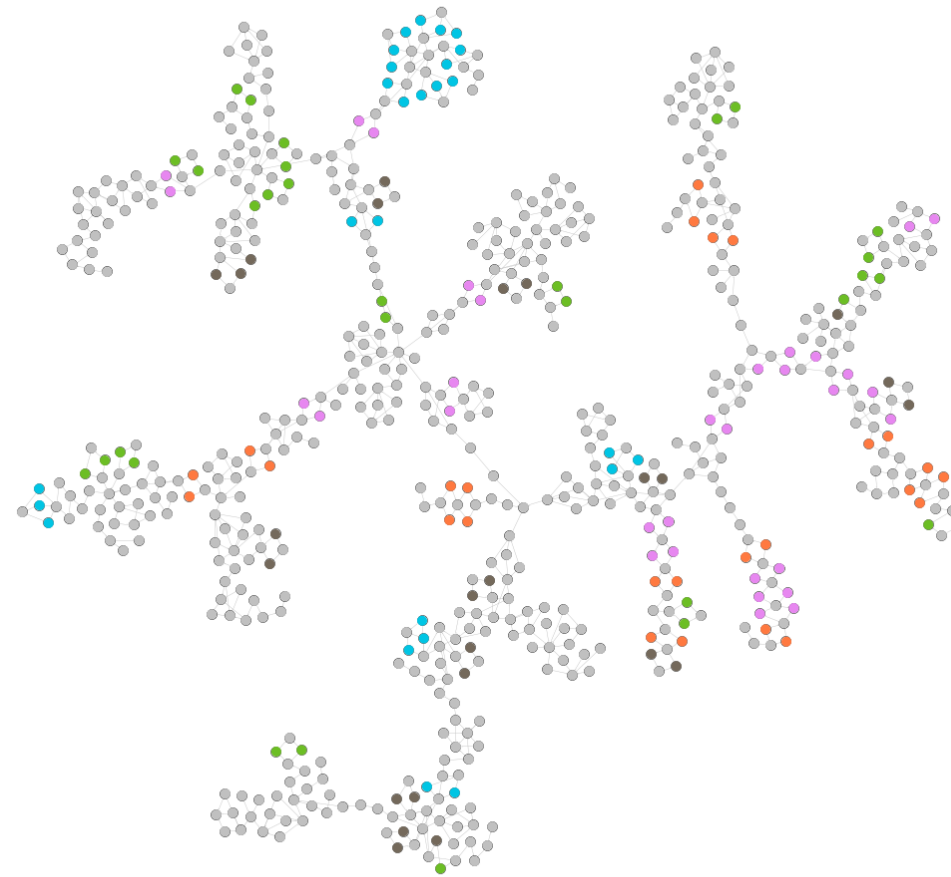


Family Network (part)

Component of 15.7 million vertices family graph
Edges = parent child relation

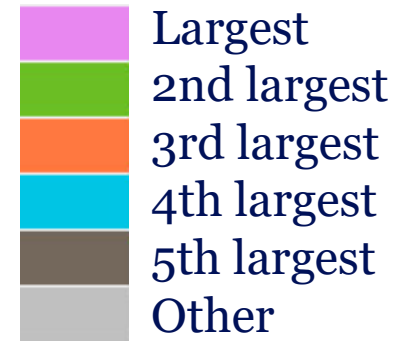


d= 1



d= 2

Equivalence
class size

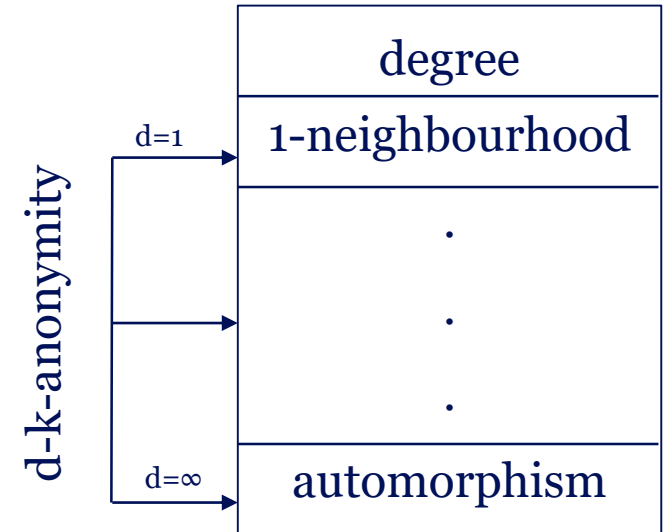


Conclusions

- **d-k-Anonymity:** a new parametrized measure for anonymity that accounts for the utility/anonymity trade-off
- For well known random graph models, most anonymity is lost at $d=2$
- We can measure anonymity of the full family network of the Netherlands

Future work

- Apply to network of neighborhood, work and school relations
- Account for network attributes (vertices, edges)
- Anonymization



Thank you!

Presentations by our team:

Sess.	D/T (EDT)		Presenter	Title
E&I	June 30	9:50	<i>Carolina Mattsson</i>	Process-driven network analysis of a mobile money system in Asia
WiNS	July 1	10:20	<i>Carolina Mattsson</i>	Real-world walk processes on networks
8	July 5	8:45	<i>Carolina Mattsson</i>	Functional structure in production networks
9	July 5	9:45	<i>Dafne van Kuppevelt</i>	Towards understanding the evolving community structure of board interlock networks
54	July 6	12:45	<i>Rachel de Jong</i>	Measuring Anonymity in Complex Networks
112	July 9	8:15	<i>Frank Takes</i>	The community structure of global scientific collaboration
151	July 10	11:30	<i>Hanjo Boekhout</i>	Investigating scientific mobility in co-authorship networks using multilayer temporal motifs

Leiden Computational Network Science Lab

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<https://theses.liacs.nl/cs> (to be published)

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